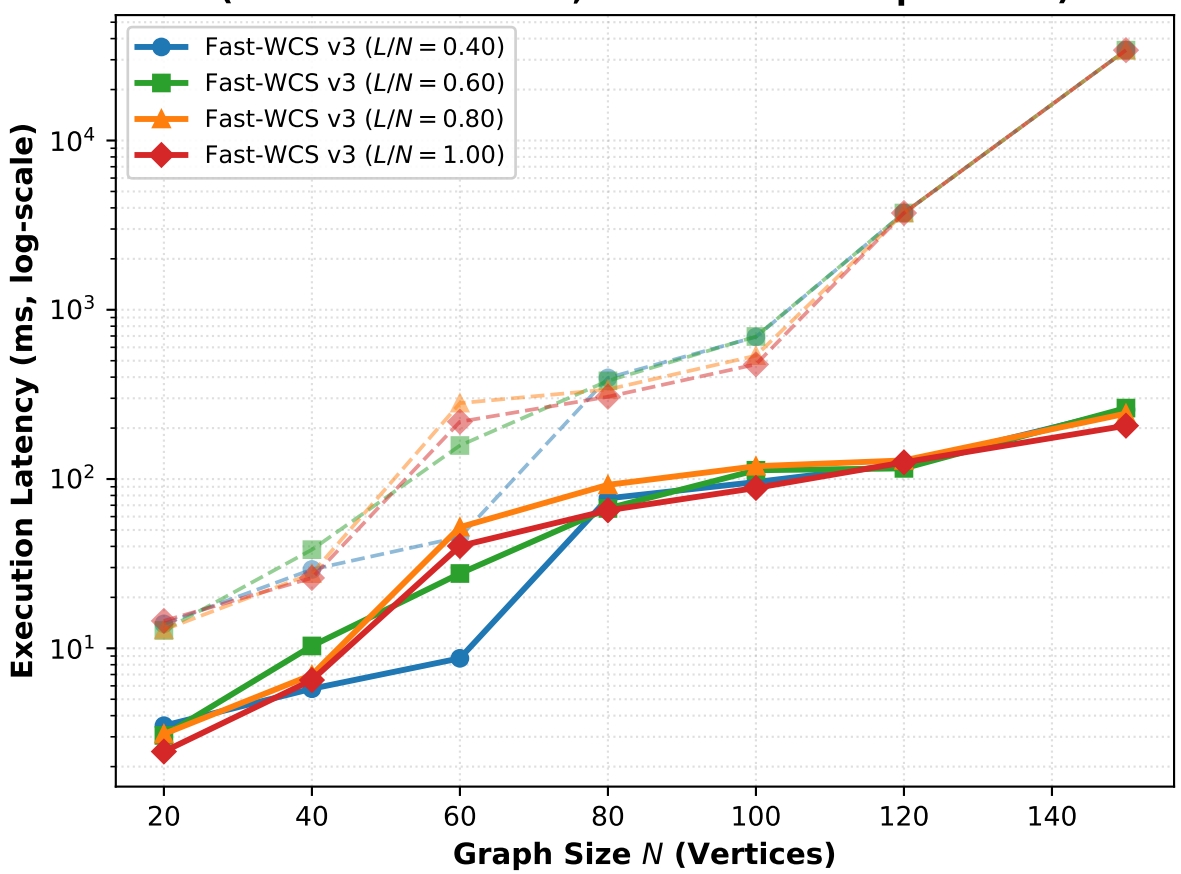
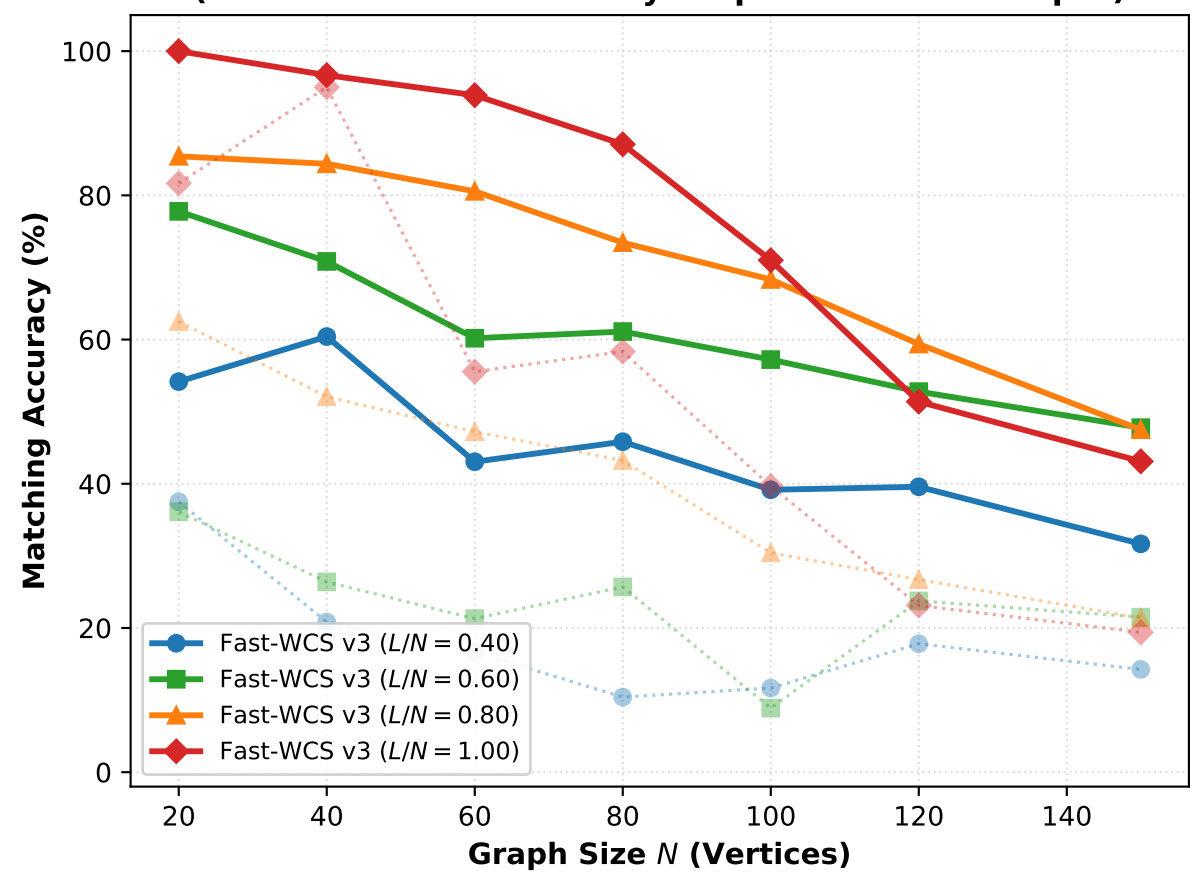
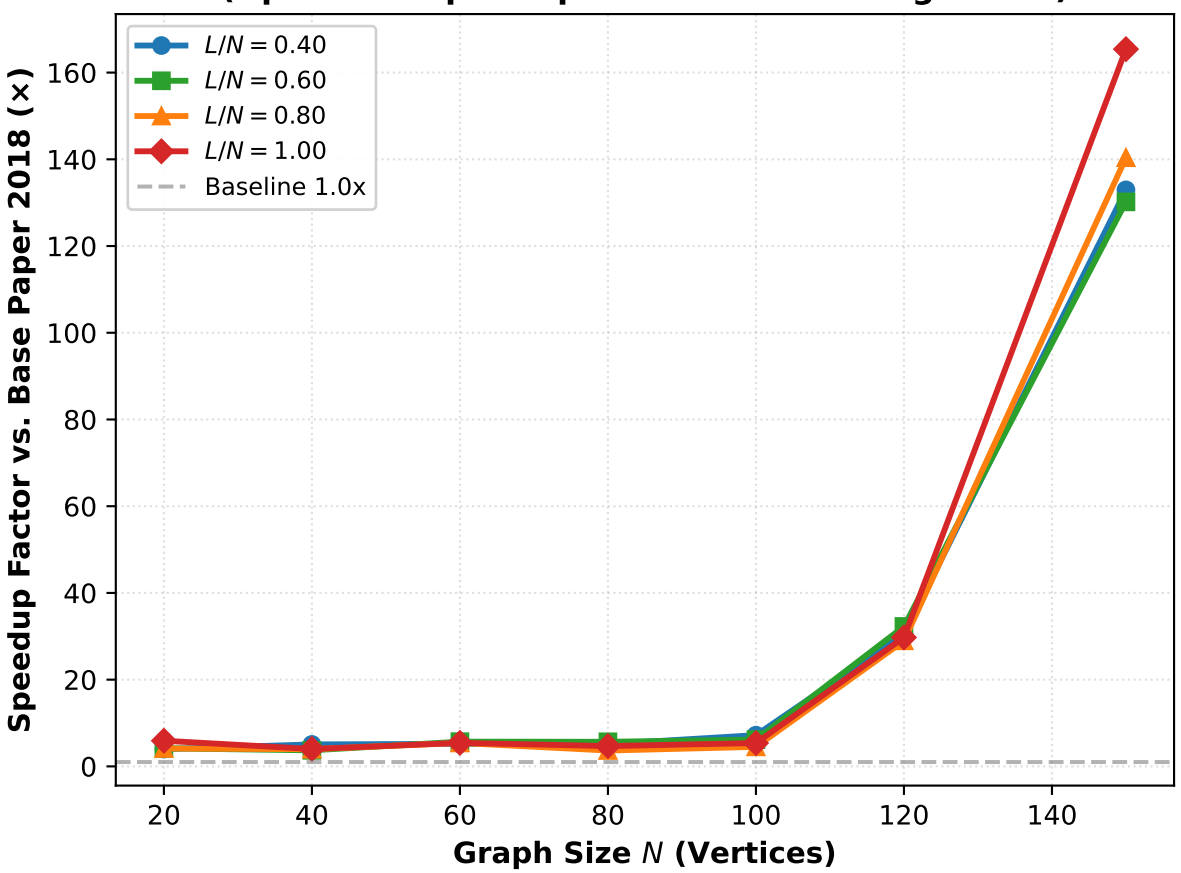
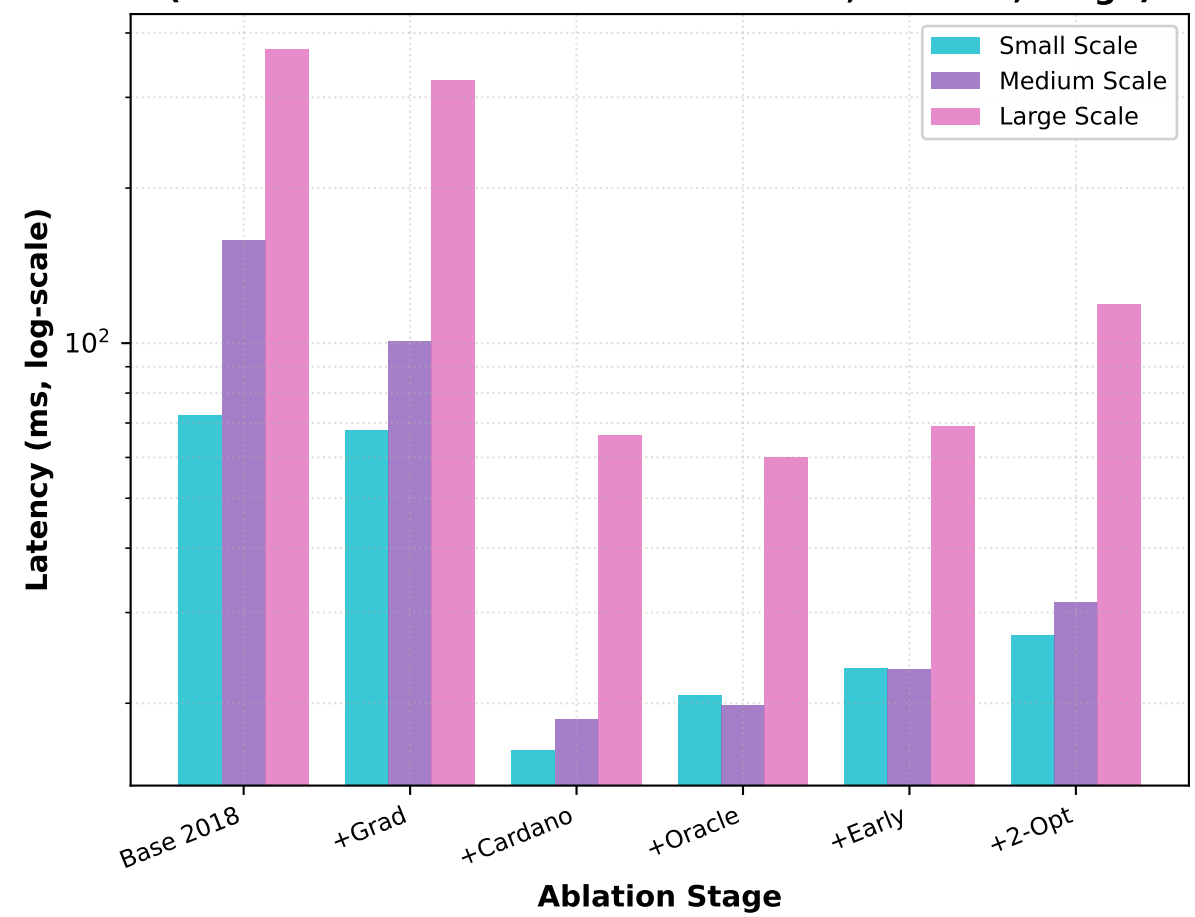

(a) Asymptotic Runtime vs. N and Subgraph Ratio L/N
(Solid: Fast-WCS v3, Dashed: Base Paper 2018)
Fast-WCS v3 (L/N = 0.40)
Fast-WCS v3 (L/N = 0.60)
Fast-WCS v3 (L/N = 0.80)
Fast-WCS v3 (L/N = 1.00)
Execution Latency (ms, log-scale)
Graph Size N (Vertices)


(b) Structural Recovery Accuracy vs. Problem Size N
(Fast-WCS v3 consistently outperforms Base Paper)
Fast-WCS v3 (L/N = 0.40)
Fast-WCS v3 (L/N = 0.60)
Fast-WCS v3 (L/N = 0.80)
Fast-WCS v3 (L/N = 1.00)
Matching Accuracy (%)
Graph Size N (Vertices)


(c) Empirical Wall-Clock Acceleration Factor
(Up to 15× speedup over 2018 base algorithm)
L/N = 0.40
L/N = 0.60
L/N = 0.80
L/N = 1.00
Baseline 1.0x
Speedup Factor vs. Base Paper 2018 (×)
Graph Size N (Vertices)


(d) Multi-Scale Component Ablation Across Problem Regimes
(Consistent acceleration across Small, Medium, Large)
Small Scale
Medium Scale
Large Scale
Latency (ms, log-scale)
Base 2018
+Grad
+Cardano
+Oracle
+Early
+2-Opt
Ablation Stage